MODELS OF ARITHMETIC OF TWO MINDS ABOUT CONSISTENCY

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Abstract. For each n we construct Turing machines $T_1, \dots, T_n$, none of which halts, and associated provability predicates $\mathrm{Pr}_{T_i}(x)$, asserting that x has a proof appearing before T_i halts. Each Pr_{T_i} is PA-provably a subpredicate of the standard provability predicate and defines exactly the standard provability relation in the standard model; in particular PA proves $\mathrm{Con(PA)} \to \mathrm{Con}_{T_i}(\mathrm{PA})$ for the associated consistency statements. We show, assuming only the consistency of PA, that these are the *only* implications: for every $S \subseteq \{1, \dots, n\}$ the theory

$$\mathrm{PA} \ + \ \neg\mathrm{Con(PA)} \ + \ \{\mathrm{Con}_{T_i}(\mathrm{PA}) : i \in S\} \ + \ \{\neg\mathrm{Con}_{T_i}(\mathrm{PA}) : i \notin S\}$$

is consistent. In particular, a single model of PA can satisfy $\mathrm{Con}_{T_1}(\mathrm{PA}) \wedge \neg\mathrm{Con}_{T_2}(\mathrm{PA})$: whether a model of arithmetic "thinks PA is consistent" is not an invariant of the model, but depends on the chosen arithmetization of provability. The proof combines Rosser's witness-comparison method with the recursion theorem, relocating the witness comparison from the syntax of a sentence into the dynamics of a machine. We also prove a normal form theorem: for every machine T, PA proves that $\mathrm{Con}_T(\mathrm{PA})$ is equivalent to the disjunction of Con(PA) with the assertion that T halts before any inconsistency proof appears. Using it, we reduce the classification of the machine consistency statements up to provable equivalence to two sharply posed questions about Σ_1 sentences under witness comparison. The main results have been formally verified in Lean 4.

1. Introduction

Gödel's second incompleteness theorem asserts that a consistent, sufficiently strong, effectively axiomatized theory such as PA neither proves nor refutes its own consistency statement Con(PA), and the completeness theorem then yields models of PA satisfying Con(PA) alongside models satisfying $\neg$Con(PA). In a loose way of speaking, one is tempted to say that some models of PA *think* PA is consistent while others think it is not, as though this were a property of the model. But not so fast. The sentence Con(PA) is not handed to us by the theory; it is manufactured from choices—a Gödel numbering, a formalized proof predicate, an enumeration of the axioms—and the independence phenomenon is notoriously sensitive to these choices. The purpose of this note is to make the sensitivity vivid in a sharp form: we produce a *single* model of PA which is of two minds about the consistency of PA, satisfying one entirely natural-looking arithmetization of "PA is consistent" while refuting another, where both arithmetizations define exactly the true provability relation in the standard model. More generally, we classify completely the possible patterns of consistency verdicts for a natural family of such arithmetizations.

That the consistency statement is intensional is of course a classical insight. Rosser's theorem [10] already provides a one-line witness: for the Rosser provability predicate

$$\mathrm{Pr}^R(x) \ := \ \exists y \left(\mathrm{Proof}(y, x) \wedge \forall z \le y \, \neg\mathrm{Proof}(z, \ulcorner 0{=}1 \urcorner)\right),$$

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the associated consistency statement is an outright theorem of PA, so that any model of PA + $\neg$Con(PA) already satisfies consistency-according-to-Rosser while refuting consistency-according-to-Hilbert–Bernays—and the two predicates are extensionally identical in the standard model (assuming, as we do throughout, that PA is in fact consistent). Feferman's fundamental study [1] showed that varying the *enumeration of the axioms* produces similar effects, and isolated conditions on an arithmetization under which the second incompleteness theorem is recovered; the derivability conditions of Hilbert–Bernays and Löb play the same role on the side of the proof predicate. There is by now a rough consensus among logicians that an acceptable formalization of provability ought to be provably equivalent to the standard one, precisely because predicates violating this permit the effects above; on this view the deviant predicates do not *genuinely express* provability, and the second incompleteness theorem is not threatened. See Halbach and Visser [4, 5] for a careful contemporary analysis of the axes of intensionality, and [6] for striking effects obtained purely on the axis of axiom enumeration.

We take no issue with that consensus; our point is that the consensus does not dissolve the descriptive question but rather poses it. Grant that the deviant predicates are deviant. They are nevertheless *extensionally correct*: in the real world—in the standard model—they define the very same relation "y is a proof of x" as the standard predicate, and the ones constructed here are even PA-provably *sound* subpredicates of the standard one, satisfying necessitation, and only ever *undercounting* proofs. How wildly can the consistency verdicts of such predicates disagree, and how does the disagreement distribute across the models of PA? The answer, it turns out, is: as wildly as is not outright provably impossible. We regard this as intensionality studied as *geography* rather than as pathology—a complete map of where the verdicts can land—and the second incompleteness theorem itself supplies the engine of the construction.

Our arithmetizations arise from a machine cutoff. For a Turing machine T which (in fact) never halts, let $\mathrm{Pr}_T(x)$ assert that x has a proof appearing strictly before T halts, that is, a proof coded by some y such that T has not halted within y steps. Since T never really halts, Pr_T defines the standard provability relation in the standard model; and provably in PA, whatever Pr_T counts as a proof, so does the standard predicate. In a model of PA + $\neg$Con(PA), however, the verdict of the associated consistency statement $\mathrm{Con}_T(\mathrm{PA})$ is controlled by a race between two possibly nonstandard numbers: the least proof of 0=1, and the halting time of T, if any. The main theorem states that this race can be rigged in every conceivable way, simultaneously for any finite number of machines.

Theorem 1.1 (Main Theorem; see Theorem 4.1)**.** *Assume* PA *is consistent, and let* $n \geq 1$*. There exist Turing machines* $T_1, \ldots, T_n$*, none of which halts, such that:*

1. *each* Pr_{T_i} *defines the standard provability relation in the standard model, and* $\mathrm{PA} \vdash \mathrm{Pr}_{T_i}(x) \to \mathrm{Pr}(x)$*, so in particular* $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$*;*
2. *for every* $S \subseteq \{1, \ldots, n\}$*, the theory*

$$\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) + \{\mathrm{Con}_{T_i}(\mathrm{PA}) : i \in S\} + \{\neg\mathrm{Con}_{T_i}(\mathrm{PA}) : i \notin S\}$$

is consistent.

Consequently, the implications $\mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$ *generate all provable implications among the sentences* $\mathrm{Con}(\mathrm{PA}), \mathrm{Con}_{T_1}(\mathrm{PA}), \ldots, \mathrm{Con}_{T_n}(\mathrm{PA})$*, and the realizable verdict patterns in models of* PA *are exactly: all verdicts true, or* Con(PA) *false and the remaining verdicts arbitrary.*

Taking $n = 2$ and $S = \{1\}$ yields the promised bifurcation.

Corollary 1.2. *Assume* PA *is consistent. There are two arithmetizations of provability, each defining the standard provability relation in the standard model and each* PA*-provably a subpredicate of the standard predicate, together with a single model* $M \models \mathrm{PA}$*, such that* M *satisfies the consistency of* PA *as formulated with the first and refutes it as formulated with the second.*

The proof is a Rosser argument, but with a twist we found illuminating. In Rosser's construction the witness comparison—"my proof comes before any refutation"—is written into the *syntax* of the sentence. Here the comparison is relocated into the *dynamics* of the machines: by the recursion theorem, the machines $T_1, \ldots, T_n$ jointly watch for a PA-proof that some verdict pattern is impossible, and upon finding one, they conspire to *instantiate* exactly that pattern, each machine halting or entering an eternal loop according to its role in it. Any proof constraining the pattern thereby defeats itself. No independence hypotheses on the machines are needed; the construction discharges, rather than assumes, the independence conditions posited in an earlier sketch of this result (see Remark 4.4). The technique belongs to the witness-comparison tradition systematized by Guaspari and Solovay [3], and the resulting consistency statements $\mathrm{Con}_T(\mathrm{PA})$ are Π_1 sentences strictly intermediate between the provable and Con(PA), cousins of the slow consistency statement of Friedman, Rathjen and Weiermann [2] and of Pudlák's partial consistency statements [9], with a self-referentially timed rather than proof-theoretically graded cutoff. We discuss these connections, and the precise fate of the derivability conditions, in Section 5. In Section 6 we turn from constructing particular machine consistency statements to classifying all of them: a normal form theorem (Proposition 6.2) identifies the sentences $\mathrm{Con}_T(\mathrm{PA})$, up to PA-provable equivalence, with the disjunctions of Con(PA) with sentences asserting a Rosser-style victory of a machine over inconsistency, and the classification problem then resolves into two questions about Σ_1 sentences under witness comparison, which we pose but do not settle.

Everything below assumes only the consistency of PA; no soundness or ω-consistency assumptions are used,[1] and nothing is special about PA beyond what is needed for the usual coding (Remark 5.4).

The main results of this paper — Theorem 4.1 in full (clauses (1)–(3), including the necessitation and extensional-correctness parts of clause (1)), Corollary 4.2, Proposition 6.2, and Proposition 6.3 in the stage-formula form of Remark 6.4 — have been formally verified in Lean 4 over the Foundation library, with no unproved obligations and the standard axiom footprint. The verified development takes the purely syntactic form described in Remark 4.3, and is carried out over an arbitrary Δ_1-axiomatized recursively enumerable extension of $\mathsf{I}\Sigma_1$; the specialization of the hypotheses to PA itself additionally invokes the library's stated-but-unproved instance of the standard fact that PA is Δ_1-axiomatized. The verification artifact is available at `https://github.com/DavidVFeldman/two-minds-consistency` and archived at `https://doi.org/10.5281/zenodo.21881585`.

2. Machine-cutoff provability predicates

Convention 2.1. Fix a standard Gödel numbering and a standard proof predicate $\mathrm{Proof}(y, x)$ for PA, with $\mathrm{Pr}(x) := \exists y\, \mathrm{Proof}(y, x)$ and $\mathrm{Con}(\mathrm{PA}) := \neg\mathrm{Pr}(\ulcorner 0{=}1 \urcorner)$, satisfying the usual properties: Proof is Δ_1 in PA, each proof codes a sequence of formulas with a uniquely determined conclusion (a convenience only; see the tie-break in the proof of Theorem 4.1), and the Hilbert–Bernays–Löb derivability conditions hold for Pr. For a Turing machine T (with a fixed standard

[1] With one scoping caveat, which the formalization mentioned below makes precise: clauses (1)–(3) of Theorem 4.1 use only the consistency of PA (this is machine-verified). The final classification clause additionally uses the consistency of $\mathrm{PA} + \mathrm{Con}(\mathrm{PA})$ — equivalently $\mathrm{PA} \nvdash \neg\mathrm{Con}(\mathrm{PA})$ — which follows from the arithmetical soundness of PA but not from its consistency alone.

coding of machines and computations), let $\mathrm{H}_T(y)$ denote the formula "T, run on empty input, halts within y steps." We arrange the coding so that, provably in PA: $\mathrm{H}_T(y)$ is Δ_1; H_T is monotone, i.e. $\mathrm{H}_T(y) \wedge y \leq z \to \mathrm{H}_T(z)$; and computations are provably deterministic. For standard numerals these are the usual facts about formalized finite simulations. In fact all that is ever used of H_T below is that it is a provably monotone Δ_1 predicate of y; the machine, the determinism, and the halting-state bookkeeping are presentational, a point the formalization exploits (Remark 4.3).

Definition 2.2. For a Turing machine T, define

$$\mathrm{Pr}_T(x) \;:=\; \exists y\,\big(\mathrm{Proof}(y,x) \wedge \neg \mathrm{H}_T(y)\big), \qquad \mathrm{Con}_T(\mathrm{PA}) \;:=\; \neg \mathrm{Pr}_T(\ulcorner 0{=}1 \urcorner).$$

Thus Pr_T counts exactly those proofs which appear before T halts, and all proofs if T never halts. Note that $\mathrm{Con}_T(\mathrm{PA})$ is PA-provably equivalent to the Π_1 sentence $\forall y\,(\mathrm{Proof}(y, \ulcorner 0{=}1 \urcorner) \to \mathrm{H}_T(y))$.

Lemma 2.3. *For every machine T:*

1. $\mathrm{PA} \vdash \forall x\,(\mathrm{Pr}_T(x) \to \mathrm{Pr}(x))$*; hence* $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_T(\mathrm{PA})$.
2. *If T never halts, then* Pr_T *and* Pr *define the same relation in the standard model* $\mathbb{N}$*, namely true provability in* PA*; moreover, for every standard sentence φ, if* $\mathrm{PA} \vdash \varphi$ *then* $\mathrm{PA} \vdash \mathrm{Pr}_T(\ulcorner \varphi \urcorner)$ *(necessitation).*

Proof. (1) is immediate from the definition. For (2): if T never halts, then for each standard y the sentence $\neg \mathrm{H}_T(\bar{y})$ is a true Δ_1 statement about a finite simulation, hence provable in PA; so in $\mathbb{N}$ the clause $\neg \mathrm{H}_T(y)$ is vacuously satisfied and Pr_T coincides with Pr, while if $\mathrm{PA} \vdash \varphi$ with proof coded by the standard number y, then $\mathrm{PA} \vdash \mathrm{Proof}(\bar{y}, \ulcorner \varphi \urcorner) \wedge \neg \mathrm{H}_T(\bar{y})$, giving $\mathrm{PA} \vdash \mathrm{Pr}_T(\ulcorner \varphi \urcorner)$. ☐

Lemma 2.3(1) already settles half of the classification: in any model of $\mathrm{PA} + \mathrm{Con}(\mathrm{PA})$, every verdict $\mathrm{Con}_T(\mathrm{PA})$ comes out true. All the freedom, if there is any, lives inside the models of $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$, where it is governed by a race between the least proof of $0{=}1$ and the halting of T. The next section makes the race explicit.

3. Pattern sentences

Fix machines $T_1, \ldots, T_n$. For a set $S \subseteq \{1, \ldots, n\}$, define the *pattern sentence*

$$\sigma_S \;:=\; \exists p\,\Big(\mathrm{Proof}(p, \ulcorner 0{=}1 \urcorner) \;\wedge\; \forall q < p\, \neg \mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner) \;\wedge\; \bigwedge_{i \in S} \mathrm{H}_{T_i}(p) \;\wedge\; \bigwedge_{i \notin S} \neg \mathrm{H}_{T_i}(p)\Big),$$

asserting that there is a least proof p of $0{=}1$, and that exactly the machines T_i with $i \in S$ have halted by stage p.

Lemma 3.1 (Reformulation Lemma)**.** *For each $S \subseteq \{1, \ldots, n\}$,*

$$\mathrm{PA} \;\vdash\; \sigma_S \;\longleftrightarrow\; \Big(\neg \mathrm{Con}(\mathrm{PA}) \wedge \bigwedge_{i \in S} \mathrm{Con}_{T_i}(\mathrm{PA}) \wedge \bigwedge_{i \notin S} \neg \mathrm{Con}_{T_i}(\mathrm{PA})\Big).$$

Consequently, the theory $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) + \{\mathrm{Con}_{T_i}(\mathrm{PA}) : i \in S\} + \{\neg \mathrm{Con}_{T_i}(\mathrm{PA}) : i \notin S\}$ *is consistent if and only if* $\mathrm{PA} \nvdash \neg \sigma_S$.

Proof. Work in PA. Suppose σ_S, witnessed by the least inconsistency proof p. Certainly $\neg\mathrm{Con}(\mathrm{PA})$. For $i \in S$: given any y with $\mathrm{Proof}(y, \ulcorner 0{=}1 \urcorner)$, minimality gives $p \leq y$, and monotonicity of H_{T_i} gives $\mathrm{H}_{T_i}(y)$; as y was arbitrary, $\mathrm{Con}_{T_i}(\mathrm{PA})$ holds in the Π_1 form of Definition 2.2. For $i \notin S$: the witness p itself satisfies $\mathrm{Proof}(p, \ulcorner 0{=}1 \urcorner) \wedge \neg \mathrm{H}_{T_i}(p)$, so $\mathrm{Pr}_{T_i}(\ulcorner 0{=}1 \urcorner)$, i.e. $\neg\mathrm{Con}_{T_i}(\mathrm{PA})$.

Conversely, suppose $\neg\mathrm{Con}(\mathrm{PA})$ together with the displayed pattern of verdicts. By $\neg\mathrm{Con}(\mathrm{PA})$ and the least number principle there is a least inconsistency proof p. For $i \in S$: $\mathrm{Con}_{T_i}(\mathrm{PA})$ applied to p yields $\mathrm{H}_{T_i}(p)$. For $i \notin S$: $\neg\mathrm{Con}_{T_i}(\mathrm{PA})$ provides some y with $\mathrm{Proof}(y, \ulcorner 0{=}1 \urcorner) \wedge \neg\mathrm{H}_{T_i}(y)$; minimality gives $p \leq y$, and monotonicity (contrapositively) gives $\neg\mathrm{H}_{T_i}(p)$. So p witnesses σ_S.

The final claim is the completeness theorem plus the observation that the displayed theory is PA-provably equivalent to $\mathrm{PA} + \sigma_S$. □

The pattern sentences are, we would suggest, the real objects of study here: the classification problem for verdict patterns is exactly the question of which $\neg\sigma_S$ are provable, and the machines will now be built so that none of them is.

4. The construction and the main theorem

Theorem 4.1. *Assume* PA *is consistent, and let* $n \geq 1$. *There exist Turing machines* $T_1, \dots, T_n$, *none of which halts, such that* $\mathrm{PA} \nvdash \neg\sigma_S$ *for every* $S \subseteq \{1, \dots, n\}$. *Consequently, by Lemmas 2.3 and 3.1:*

1. *each* Pr_{T_i} *defines true* PA*-provability in* $\mathbb{N}$, *is* PA*-provably a subpredicate of* Pr, *and satisfies necessitation;*
2. $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$ *for each* i*; and*
3. *for every* $S \subseteq \{1, \dots, n\}$ *the theory*
$$\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) + \{\mathrm{Con}_{T_i}(\mathrm{PA}) : i \in S\} + \{\neg\mathrm{Con}_{T_i}(\mathrm{PA}) : i \notin S\}$$
is consistent.

In particular, the only provable implications among $\mathrm{Con}(\mathrm{PA}), \mathrm{Con}_{T_1}(\mathrm{PA}), \dots, \mathrm{Con}_{T_n}(\mathrm{PA})$ *are those generated by the implications* $\mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$.

Proof. The construction. By the recursion theorem (in its standard n-fold form, or by applying the ordinary recursion theorem to a single controller index from which the n machines are uniformly computed), we may construct machines $T_1, \dots, T_n$, each with access to the full tuple of indices, behaving as follows. Each T_i performs the same synchronized search, scanning $y = 0, 1, 2, \dots$ in order and checking whether y codes a PA-proof whose conclusion is one of the 2^n *watched sentences* $\neg\sigma_S$, $S \subseteq \{1, \dots, n\}$ (these sentences are computed from the indices, and are pairwise distinct; if a single stage codes proofs of several watched sentences, the trigger adopts the least pattern among them, in some fixed ordering of the patterns — this tie-break makes the construction independent of any uniqueness convention for conclusions). If the search finds a least trigger y_0, with adopted pattern S^*, then:

- if $i \in S^*$, machine T_i halts immediately;
- if $i \notin S^*$, machine T_i enters a designated absorbing loop state, in which it remains forever.

If the search never triggers, the machines search forever. Thus the family reads the first metamathematical verdict about itself and conspires to instantiate exactly the pattern just "refuted." We arrange the loop state explicitly so that PA proves, by induction, that a machine in the loop state at some stage is in it at every later stage and never halts.

No watched sentence is provable. Suppose toward a contradiction that $\mathrm{PA} \vdash \neg\sigma_S$ for some S; then the search really triggers, say least at y_0 (a standard number), with adopted pattern S^* (which need not be the S we started from — the contradiction below runs through S^*). Note that $\neg\sigma_{S^*}$ is really provable, y_0 coding a proof of it. The subsequent behavior of each machine is a concrete finite computational fact: there is a standard number t, computable from y_0, such that each T_i with $i \in S^*$ halts within t steps, and each T_i with $i \notin S^*$ enters the loop state within t steps. Being true Δ_1 statements about finite simulations, these facts are provable:

$$\mathrm{PA} \vdash \mathrm{H}_{T_i}(\bar{t}) \quad (i \in S^*), \qquad \mathrm{PA} \vdash \forall y\, \neg \mathrm{H}_{T_i}(y) \quad (i \notin S^*),$$

the latter by combining the provable finite fact that T_i is in the loop state at stage $\bar{t}$ with the provable absorption of the loop state.

Now reason inside $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$. By the least number principle there is a least inconsistency proof p. Suppose first $p > \bar{t}$. Then for $i \in S^*$, monotonicity and $\mathrm{H}_{T_i}(\bar{t})$ give $\mathrm{H}_{T_i}(p)$; and for $i \notin S^*$ we have $\neg\mathrm{H}_{T_i}(p)$ outright. So p witnesses σ_{S^*}—contradicting the PA-provable $\neg\sigma_{S^*}$. Hence $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) \vdash p \leq \bar{t}$, and so

$$\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) \ \vdash\ \exists q \leq \bar{t}\, \mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner).$$

But t is standard and PA is (really) consistent, so for each standard $q \leq t$ the sentence $\neg\mathrm{Proof}(\bar{q}, \ulcorner 0{=}1 \urcorner)$ is a true Δ_1 sentence, hence PA-provable; conjoining these finitely many facts, $\mathrm{PA} \vdash \forall q \leq \bar{t}\, \neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)$. Therefore $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$ is inconsistent, so $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA})$—contradicting the second incompleteness theorem, which applies since PA is consistent.

Conclusion. No watched sentence is provable; hence in the real world the search never triggers, no T_i ever halts, and Lemma 2.3 yields (1) and (2). For (3), since $\mathrm{PA} \nvdash \neg\sigma_S$, the theory $\mathrm{PA} + \sigma_S$ is consistent, and Lemma 3.1 identifies it with the displayed theory. For the final claim: any purported provable implication not generated by $\mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$ would exclude one of the verdict patterns shown consistent in (2) and (3), or a pattern with all verdicts true, which holds in any model of $\mathrm{PA} + \mathrm{Con}(\mathrm{PA})$. □

Corollary 4.2 (A model of two minds)**.** *Assume* PA *is consistent. With* $n = 2$ *and* $S = \{1\}$ *in Theorem 4.1, there is a model* $M \models \mathrm{PA}$ *with*

$$M \models \mathrm{Con}_{T_1}(\mathrm{PA}) \wedge \neg\mathrm{Con}_{T_2}(\mathrm{PA}),$$

although Pr_{T_1} *and* Pr_{T_2} *both define true* PA*-provability in the standard model, both are* PA*-provably sound subpredicates of the standard provability predicate, and both satisfy necessitation. Whether a model of* PA *"thinks* PA *is consistent" is thus not a property of the model alone.*

Remark 4.3 (The syntactic reading)*.* The construction admits an equivalent, purely syntactic description, which is the form taken by the machine-checked proof. By the simultaneous diagonal lemma there are sentences δ_S $(S \subseteq \{1, \ldots, n\})$, pairwise distinct, with $\mathrm{PA} \vdash \delta_S \leftrightarrow \neg\sigma_S$, where the stage predicates standing in for the H_{T_i} are explicit, non-self-referential Δ_1 formulas reading the codes $\ulcorner \delta_{S'} \urcorner$ as fixed numerals: "some stage $y_0 \leq y$ is least among stages coding a proof of some $\delta_{S'}$, and the least pattern proved at y_0 contains i." The witness comparison, which the machine formulation relocates from syntax into dynamics, is thereby relocated back: the family (δ_S) is a simultaneous Rosser-style fixed point, each member asserting that its own pattern is not the one instantiated at the least trigger, and no recursion theorem beyond the diagonal lemma is needed. The two descriptions are intertranslatable; we find the dynamical one the more vivid and the syntactic one the more economical to verify.

Remark 4.4 (Nonstandard dynamics; independence discharged)*.* An earlier sketch of this result posited, as hypotheses, the mutual independence of the halting statements of the machines and their joint independence from Con(PA), and then faced the problem—correctly identified in that sketch—that Boolean independence cannot by itself control the *relative position* of nonstandard witnesses: one needs T_i to halt *below* the least inconsistency proof, and no combination of independence hypotheses delivers an order-type. The construction above dissolves the problem rather than solving it: the machines are not chosen independent, they are chosen self-referential, so that any proof constraining the verdict pattern defeats itself, and the positional control is exactly what the pattern sentences σ_S assert.

It is worth pausing over what the machines actually do inside a model $M \models \mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$. Since every sentence is M-provable there, each watched sentence has a (nonstandard) proof in M, so from M's internal point of view the search *does* trigger, at some nonstandard stage, and the machines then halt or loop according to the M-least trigger's pattern S^*_M. The theorem says that the model may be chosen so that this internal drama plays out in any prescribed position relative to the least inconsistency proof. In the real world, meanwhile, the search runs forever and nothing ever happens: the machines are, one might say, waiting for a contradiction that never comes.

5. Remarks, connections, and questions

Remark 5.1 (The derivability conditions, and how the escape works). Each Pr_{T_i} satisfies the first derivability condition (necessitation, Lemma 2.3), but must fail the second: a PA-verified closure of Pr_{T_i} under modus ponens would, with necessitation and provable Σ_1-completeness for Pr_{T_i}, put Pr_{T_i} back under the hypotheses of the second incompleteness theorem in Löb's form, and the analysis below shows $\mathrm{Con}_{T_i}(\mathrm{PA})$ behaves incompatibly with that. Concretely the failure is visible: modus ponens applied to proofs below the cutoff produces a longer proof, which the cutoff may decapitate. The third condition (provable Σ_1-completeness *for* Pr_{T_i} *itself*) fails for the same reason: certifying a Pr_{T_i}-fact requires a proof appearing in time. It is worth contrasting the shape of the escape with Rosser's: Rosser's consistency statement becomes an outright theorem of PA, whereas our $\mathrm{Con}_{T_i}(\mathrm{PA})$ remains unprovable—indeed $\mathrm{PA} + \neg\mathrm{Con}_{T_i}(\mathrm{PA})$ is consistent by Theorem 4.1—and unrefutable, even over $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$. The deviation from the standard predicate manifests not as a collapse of the consistency statement into provability but as its detachment from $\neg\mathrm{Con}(\mathrm{PA})$: the standard verdict no longer decides the deviant one.

Remark 5.2 (Intermediate consistency statements). Each $\mathrm{Con}_{T_i}(\mathrm{PA})$ is a Π_1 sentence with $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_{T_i}(\mathrm{PA})$, $\mathrm{PA} \nvdash \mathrm{Con}_{T_i}(\mathrm{PA})$, and $\mathrm{PA} + \mathrm{Con}_{T_i}(\mathrm{PA}) \nvdash \mathrm{Con}(\mathrm{PA})$: a nontrivial consistency-like statement strictly intermediate between the provable and full $\mathrm{Con}(\mathrm{PA})$. In spirit, $\mathrm{Con}_T(\mathrm{PA})$ is a partial or cut-off consistency statement—"there is no inconsistency proof of size below the halting time of T"—placing it in the company of Pudlák's finite and restricted consistency statements [9] and of the slow consistency of Friedman, Rathjen, and Weiermann [2] (see also Henk and Pakhomov [7]), but with a cutoff timed self-referentially by the recursion theorem rather than graded by a proof-theoretic hierarchy. It would be interesting to locate the $\mathrm{Con}_T(\mathrm{PA})$ of Theorem 4.1 in the interpretability degrees alongside those statements.

Remark 5.3 (Other axes of intensionality). The construction here varies the proof predicate while holding the Gödel numbering and axiom enumeration fixed. Feferman [1] showed how much can be achieved by varying the enumeration of axioms alone; a modern instance of that trick, in which a PA-enumeration is arranged to acquire the consistency strength of large cardinal hypotheses, appears in [6]. Halbach and Visser [4, 5] taxonomize the axes systematically. It seems fair to describe the present contribution, against that background, as: the machine-cutoff formulation; the complete classification of verdict patterns for finitely many extensionally correct, provably sound subpredicates; and the resulting single-model bifurcation phenomenon of Corollary 4.2.

Remark 5.4 (Base theory). We have stated everything for PA for concreteness, but the argument uses only: Δ_1 representation of the relevant predicates with provable monotonicity, the least number principle for Δ_1 formulas, provable Σ_1-completeness for the standard predicate at standard instances, the diagonal lemma, and the second incompleteness theorem. Accordingly, the machine-checked development mentioned in the introduction is carried out over an arbitrary Δ_1-axiomatized recursively enumerable theory extending $\mathsf{I}\Sigma_1$, and Theorem 4.1, Corollary 4.2, and Proposition 6.2 hold at that generality; the statements for PA are instances.

Further questions appear at the end of the next section.

6. Which sentences are machine consistency statements?

Question: which sentences θ arise, up to PA-provable equivalence, as $\mathrm{Con}_T(\mathrm{PA})$ for a machine T that never halts? By Lemma 2.3, $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \theta$ is necessary, and one might naively conjecture it sufficient, which would make the machine consistency statements a normal form for all Π_1 consequences of consistency. The situation is more delicate and, we think, more interesting. The key is to make explicit the Rosser-style race that has been implicit throughout.

Definition 6.1. For a machine T, let $\mathrm{HaltAt}_T(s)$ be the Δ_1 formula asserting that T halts exactly at step s (provably, such s is unique if it exists), and define

$$\mathrm{Halt}^R(T) \;:=\; \exists s \left(\mathrm{HaltAt}_T(s) \wedge \forall q < s\, \neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)\right),$$

a Σ_1 sentence asserting that T *halts before any inconsistency proof appears*—that T wins the Rosser race against inconsistency. For a never-halting T, $\mathrm{Halt}^R(T)$ is false in $\mathbb{N}$.

Proposition 6.2 (Normal form)**.** *For* every *machine T,*

$$\mathrm{PA} \;\vdash\; \mathrm{Con}_T(\mathrm{PA}) \;\longleftrightarrow\; \mathrm{Con}(\mathrm{PA}) \vee \mathrm{Halt}^R(T).$$

Equivalently, with $\sigma_{\{1\}}$ the pattern sentence of Section 3 for the single machine T: $\mathrm{PA} \vdash \sigma_{\{1\}} \leftrightarrow \neg\mathrm{Con}(\mathrm{PA}) \wedge \mathrm{Halt}^R(T)$.

Proof. Work in PA. ($\leftarrow$) $\mathrm{Con}(\mathrm{PA}) \to \mathrm{Con}_T(\mathrm{PA})$ is Lemma 2.3. Suppose $\mathrm{Halt}^R(T)$, witnessed by s, and let y be any inconsistency proof; then $y \geq s$ by the witness clause, so $\mathrm{H}_T(y)$ by monotonicity, giving $\mathrm{Con}_T(\mathrm{PA})$ in its Π_1 form. ($\to$) Suppose $\mathrm{Con}_T(\mathrm{PA})$; if $\mathrm{Con}(\mathrm{PA})$ we are done, so suppose also $\neg\mathrm{Con}(\mathrm{PA})$, with least inconsistency proof p. Applying $\mathrm{Con}_T(\mathrm{PA})$ to p gives $\mathrm{H}_T(p)$, so T halts at some $s \leq p$; and for $q < s \leq p$, minimality of p gives $\neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)$. Thus $\mathrm{Halt}^R(T)$. □

Thus every machine consistency statement asserts exactly: *either* PA *is consistent, or the machine beat the inconsistency to the tape.* The classification problem therefore splits along the two disjuncts. Writing $[\,\cdot\,]$ for PA-provable equivalence classes, Proposition 6.2 gives

$$\{[\mathrm{Con}_T(\mathrm{PA})] : T \text{ never halts}\} \;=\; \{[\mathrm{Con}(\mathrm{PA}) \vee \rho] : \rho \in \mathcal{H}\}, \qquad \mathcal{H} := \{[\mathrm{Halt}^R(T)] : T \text{ never halts}\},$$

and $\mathcal{H}$ consists of ($\equiv$-classes of) Σ_1 sentences false in $\mathbb{N}$. The next proposition shows $\mathcal{H}$ is plentiful: every Σ_1 sentence false in $\mathbb{N}$ contributes its *Rosser truncation* to $\mathcal{H}$, at a pacing determined by a machine implementation.

Proposition 6.3 (Realization)**.** *Let $\rho = \exists x\, \delta(x)$ be a Σ_1 sentence ($\delta \in \Delta_0$, without loss of generality) false in $\mathbb{N}$. Let T_ρ be the machine that checks $\delta(0), \delta(1), \ldots$ in order and halts upon first finding a witness, and let $c(x)$ denote the* PA*-provably total, provably strictly monotone function giving the step at which T_ρ completes the check of x. Then T_ρ never halts, and*

$$\mathrm{PA} \;\vdash\; \mathrm{Halt}^R(T_\rho) \;\longleftrightarrow\; \rho^{R,c}, \qquad \textit{where} \quad \rho^{R,c} := \exists x \left(\delta(x) \wedge \forall q < c(x)\, \neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)\right).$$

Consequently $\mathrm{Con}(\mathrm{PA}) \vee \rho^{R,c}$ *is, up to provable equivalence, a machine consistency statement, namely* $\mathrm{Con}_{T_\rho}(\mathrm{PA})$.

Proof. T_ρ never halts since ρ is false in $\mathbb{N}$. Work in PA. If $\mathrm{Halt}^R(T_\rho)$ holds with witness s, then provably $s = c(x_0)$ for the least witness x_0 of δ, and the clause $\forall q < s\, \neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)$ is exactly the truncation clause at x_0; so $\rho^{R,c}$ holds. Conversely, given x with $\delta(x)$ and no inconsistency proof below $c(x)$, the least witness $x_0 \leq x$ satisfies $c(x_0) \leq c(x)$ by monotonicity, so T_ρ halts at $c(x_0)$ with no inconsistency proof below it, giving $\mathrm{Halt}^R(T_\rho)$. □

Remark 6.4 (The pacing is an artifact). The pacing function c belongs to the machine presentation, not to the mathematics. Working directly with monotone Δ_1 stage formulas $\eta(y)$ in place of machines — as the formalization does (Remark 4.3) — one may take $\eta_\rho(y) := \exists x \leq y\, \delta(x)$, and the first stage at which η_ρ fires *is* the least witness of ρ: the truncation of Proposition 6.3 lands at $c = \mathrm{id}$, i.e., at

$$\rho^R \ := \ \exists x \left(\delta(x) \wedge \forall x' < x\, \neg\delta(x') \wedge \forall q < x\, \neg\mathrm{Proof}(q, \ulcorner 0{=}1 \urcorner)\right).$$

Question 6.7 is accordingly better phrased presentation-theoretically: is every Σ_1 sentence false in $\mathbb{N}$ provably equivalent to π^R for *some* Σ_1 presentation π of it? The sensitivity of π^R to the choice of presentation is then precisely the Guaspari–Solovay phenomenon. The stage-formula form of Proposition 6.3 just described — with η_ρ in place of T_ρ and truncation at the least witness — is among the machine-verified results.

Corollary 6.5 (Reduction of the classification). *A sentence θ is* PA*-provably equivalent to* $\mathrm{Con}_T(\mathrm{PA})$ *for some never-halting T if and only if* $\mathrm{PA} \vdash \theta \leftrightarrow \mathrm{Con}(\mathrm{PA}) \vee \rho$ *for some $\rho \in \mathcal{H}$. In particular, the naive conjecture above holds if and only if both of the following do:*

(A) *every Π_1 sentence θ with* $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \theta$ *satisfies:* $\theta \wedge \neg\mathrm{Con}(\mathrm{PA})$ *is* PA*-provably equivalent to a Σ_1 sentence; and*

(B) *every Σ_1 sentence false in $\mathbb{N}$ is* PA*-provably equivalent to* $\mathrm{Halt}^R(T)$ *for some never-halting T—equivalently, is provably equivalent to a Rosser truncation $\rho^{R,c}$ of some Σ_1 sentence at some machine pacing c.*

Proof. The displayed equality of classes gives the first claim. For the second: if $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \theta$, then provably $\theta \leftrightarrow (\theta \wedge \mathrm{Con}(\mathrm{PA})) \vee (\theta \wedge \neg\mathrm{Con}(\mathrm{PA})) \leftrightarrow \mathrm{Con}(\mathrm{PA}) \vee (\theta \wedge \neg\mathrm{Con}(\mathrm{PA}))$, so θ is of the required form if and only if $\theta \wedge \neg\mathrm{Con}(\mathrm{PA})$ is equivalent to a member of $\mathcal{H}$; condition (A) supplies Σ_1-ness and condition (B) upgrades Σ_1-ness (falsity in $\mathbb{N}$ being automatic, since $\theta \wedge \neg\mathrm{Con}(\mathrm{PA})$ is false in $\mathbb{N}$ by the actual consistency of PA) to membership in $\mathcal{H}$. □

We suspect both (A) and (B) fail, so that the machine consistency statements form a proper—and, by Proposition 6.3, still remarkably rich—subclass of the Π_1 consequences of consistency; but we have not proved either failure, and we record the two halves as questions.

Question 6.6 (Obstruction A). Is there a Π_1 sentence θ with $\mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to \theta$ such that $\theta \wedge \neg\mathrm{Con}(\mathrm{PA})$ is not PA-provably equivalent to any Σ_1 sentence? The slow consistency statement of [2] seems a natural candidate: over $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})$ with least inconsistency proof p, slow consistency asserts a Π_1-flavored domination condition tying the convergence of F_{ε_0} to the position of p, with no evident Σ_1 reformulation. Preservation of Σ_1 sentences under end-extensions suggests a strategy: exhibit a model of $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) + \mathrm{Con}^{\mathrm{slow}}$ with an end-extension to a model of $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA}) + \neg\mathrm{Con}^{\mathrm{slow}}$.

Question 6.7 (Obstruction B). Is every Σ_1 sentence false in $\mathbb{N}$ provably equivalent to $\mathrm{Halt}^R(T)$ for some never-halting T — equivalently (Remark 6.4), to a Rosser truncation π^R of some Σ_1 presentation π of it? The truncation changes the sentence, in general: π^R provably implies π, but the converse asks every model's witness to appear before its inconsistency proofs, which is a genuine positional demand. The case $\mathrm{PA} \vdash \rho \to \neg\mathrm{Con}(\mathrm{PA})$ seems especially delicate:

there, $\mathrm{Halt}^R(T) \equiv \rho$ would require the machine's Rosser victory—which the machine achieves precisely by seeing *no* inconsistency proof before halting—to provably certify that inconsistency proofs exist beyond its halting time. We expect the witness-comparison methods of Guaspari and Solovay [3] (see also Lindström [8]) to be the right tools here, in either direction.

Question 6.8. Theorem 4.1 handles each finite n, with machines depending on n. Is there a single uniformly computable infinite family $(T_i)_{i\in\omega}$, none halting, realizing every "locally permitted" verdict pattern—for instance, such that for every set $A \subseteq \omega$ that is Σ_1-definable in a given model of $\mathrm{PA} + \neg\mathrm{Con}(\mathrm{PA})\ldots$ —in an appropriate formulation? Already the right statement of the infinitary classification seems delicate, since σ_S becomes infinitary.

Question 6.9. The map $T \mapsto \mathrm{Con}_T(\mathrm{PA})$ invites iteration: one may form $\mathrm{Con}_T(\mathrm{PA}{+}\mathrm{Con}_{T'}(\mathrm{PA}))$ and so on. What hierarchy results, under provability or interpretability, and how does it interact with the classical progressions of iterated consistency?

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